Use Octave for Your Statistical Tasks

This thirteenth article of the mathematical journey through open source, gives a glimpse of statistical capabilities of octave.

Statistics is all about data, probabilities, averages, deviations and random numbers. Octave shows you how to compute these easily.

Data moments

Moments mean the various kinds of averages, median, modes, etc. To understand them, let's look at a random data set, which may be generated by using any of the various distributions. Let's examine the most 'natural' normal distribution–yes, the bell-shaped one. Figure 1 shows one centred around 0 (the mean) with a spread of 3 (the standard deviation), generated using *normpdf()*, as cited below. Note that Octave has a whole set of all such functions.

```
$ octave -qf
octave:1> X = -10:0.1:10;
octave:2> Y = normpdf(X, 0, 3); # Normal Density Function
octave:3> plot(X, Y, "b*");
octave:4> xlabel("X ->");
octave:5> ylabel("Y = e^{-{x^2}/3} ->");
octave:6> title("Normal Distribution");
octave:7> grid("on");
octave:8>
```

If we take all the infinite points on this curve, we would get a mean of 0 and a standard deviation of 3. But that's not practical. So, let's pick, say 10 points at random, from it. *normrnd(m, s, 10, 1)* provides the same in a 10x1 vector, following the normal distribution of mean *m* and standard deviation *s*. And then, *mean()* and *std()* give the mean and standard deviation, respectively. *mean()* could be of three types: the ordinary (arithmetic) mean ("a"), the geometric mean ("g") and the harmonic mean ("h").

```
$ octave -qf
octave:1> Y = normrnd(0, 3, 10, 1) # m=0; s=3; 10x1 vector
Y =

  -2.33218
  -3.16866
   3.41991
  -2.27566
  -1.45403
   7.79691
   2.12849
```